

Fig. 2: Actual-size PCB layout for low-noise power supply

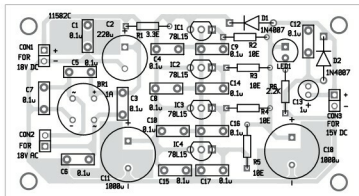


Fig. 3: Components layout of the PCB

C7, and C11. The effectiveness of the noise filtering depends on the internal resistance of the AC voltage source and diodes in the bridge.

The circuit has four 78L15 voltage regulators in parallel. Each of the regulators has an equalisation resistor (R2 through R5) at the output. The values of the equalisation resistors are equal to or greater than 10 ohms. The resistors are much higher in value than the internal output resistance of the regulators.

If required, you may add a capacitor of 0.1 μ F to each regulator (refer capacitors C4, C9, C8, C14, C10, C16, C15, and C17 in the circuit). These individual capacitors guarantee stable operation of the voltage regulators and reduce high-frequency noise to some extent.

Selection of equalisation resistors and RC filters. The output voltage of the circuit depends on the load current and equalisation resistors. A higher equalisation resistor will produce better equalisation and better filtering of the noise. But it will also lead to a higher dependency on the output voltage from the load current. That is, larger resistors will improve the filtration of the noise from the regulators but will affect load regulation. In this application, noise reduction is more important. The usual values of these resistors are from 10 to 30 ohms.

The main parameters with 10-ohm

resistors R2 through R5 are given in Table 1.

The cut-off frequency of the output filter built around the equalisation resistors and output capacitors C12, C13, and C18 should preferably be below 20Hz when we use 50Hz/60Hz AC mains input. With resistor $R = 10\text{-ohm}$ and capacitor $C = 1000\mu\text{F}$, we have frequency $F(-3\text{dB}) = 15.9\text{Hz}$ along with some harmonic suppression in the 50/60Hz AC mains present in the circuit after rectification.

Construction and testing

An actual-size PCB layout for the low-noise power supply is shown in Fig. 3 and its components layout in Fig. 4. After assembling the circuit on the PCB, connect either 18V DC across CON1 or 18V AC across CON2 as input and take 15V DC output across CON3.

This article presents a low noise power supply with four 78L15 voltage regulators in parallel that produces output of 14.4V at 0.25A. The average output current per regulator is below 75mA, which is below the defined 100mA output current for the 78L15 series. (We have an average current of 62.5mA per the regulator, given a total of $4 \times 62.5\text{mA} = 250\text{mA}$).

The circuit has a current limiting function and short-circuit protection. It does not need any adjustment to work properly.

PARTS LIST

Semiconductors:

- IC1-IC4 - 78L15, 15V regulator (TO-92 package)
- BR1 - Bridge rectifier (1A)
- LED1 - 5mm LED
- D1, D2 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 3.3-ohm
- R2-R5 - 10-ohm
- R6 - 2.2-kilo-ohm

Capacitors:

- C1, C3-C10, C12, C14-C17 - 0.1 μ F ceramic disk
- C2 - 220 μ F, 35V electrolytic
- C11 - 1000 μ F, 35V electrolytic
- C13 - 1 μ F, 25V electrolytic
- C18 - 1000 μ F, 25V electrolytic

Miscellaneous:

- CON1-CON3 - 2-pin connector

Preferably, we should use a 78L15 regulator with a tolerance of $\pm 4\%$ or better and equalisation resistors with maximal practical values. Here, we are not concerned about output regulation but are interested in the low-noise output and equal output current distribution between voltage regulators connected in parallel. **EFY**

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